

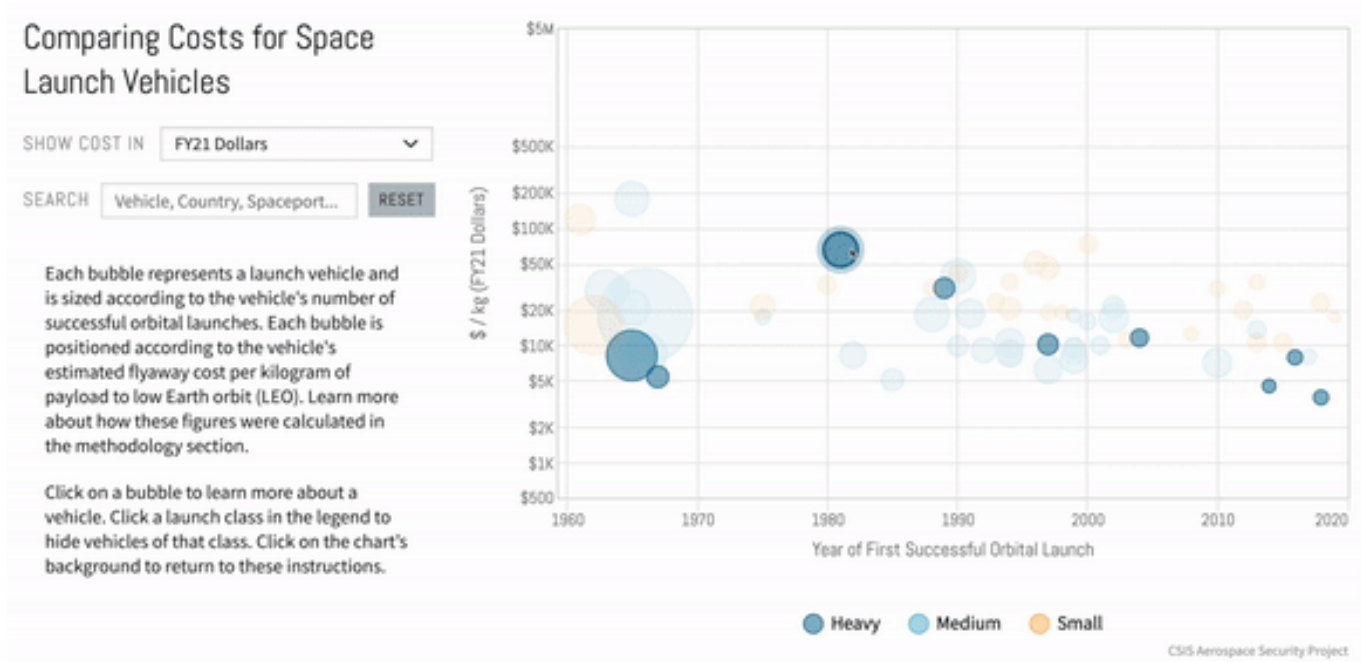
Aerospace Security

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Space Launch to Low Earth Orbit: How Much Does It Cost?

LAST UPDATED September 1, 2022

BY Thomas G. Roberts



To view and interact with our full dataset, please view on a desktop computer.

This data repository accompanies Appendix 1 of Boost-Phase Missile Defense: Interrogating the Assumptions, a featured report from the CSIS Missile Defense Project.

ALTHOUGH SPACE LAUNCH VEHICLES often have vastly different characteristics from one another—including the orbital regimes into which they can place payloads, the spaceports from which they can be

launched, and their likelihood of success or failure—they all share the same core mission: to safely place payloads into orbit around the Earth. Some critical differences between launch vehicles, like total lift capability and whether any of their components are designed to be reused, may lead to drastically different launch costs. This data repository compares costs between space launch vehicles by incorporating many vehicle characteristics into a single figure: the cost to launch one kilogram of payload mass to low Earth orbit (LEO) as part of a dedicated launch.

In the interactive chart above, use the “Show Cost In” input field to toggle between current-year dollars and then-year dollars. Selecting “FY21 Dollars” inflates cost estimates to their dollar values in fiscal year 2021. Selecting “Then-Year Dollars” shows cost estimates for vehicles at the time of their first successful orbital launch. All adjustments for inflation in this data repository are made using the GDP Chained Price Index published by the Office of Management and Budget in Historical Table 10.1. In FY21 dollars, newer launch vehicles tend to offer lower costs than older launch vehicles, with a gradual decline from 1957 to 2005, and a steeper decline between 2005 and 2020. In then-year dollars, per-kilogram costs increased from 1957 to 2005 and generally decreased from 2005 to 2020.

To learn more about how a particular vehicle’s launch costs compares to others, click on a bubble or search for a vehicle’s name in the search field. The search field can also be used to highlight launch vehicles by family, country, launch provider, or spaceport. Use the “Reset” button to remove the search query. Hover over the “Heavy,” “Medium,” or “Small” buttons in the chart’s legend to view launch vehicles of a particular payload mass class. Click one of the class buttons to remove the corresponding set of bubbles from the chart. Click and drag on the chart area to zoom into a particular subset of bubbles.

We encourage corrections, additions, and suggestions. Please direct all messages to aerospace@csis.org.

Methodology

Estimating costs for space launch vehicles is rarely straightforward. In many cases, space launches are arranged through private or classified contracts.¹ In other cases, launch providers may provide costs for a single configuration of a launch vehicle, despite offering a wide range of variants of the vehicle to potential customers with vastly different capabilities.² Most critically, the very definition of “launch cost” is subject to interpretation. In this data repository, the per-kilogram launch cost provided in the interactive chart is typically the “unit flyaway cost,” a term borrowed from the aviation industry and defined in the “Definitions” subsection of this page. For older launch vehicles, which were often directly funded by civil space agencies and military services, unit flyaway costs are not always available. In those

cases, non-recurring costs, such as research and development, may be included as part of the figure. Due to these discrepancies, the data source is provided in the interactive chart on a vehicle-by-vehicle basis. Full citations can be found in the “Sources” section at the bottom of this page.

Many of the cited sources directly provide cost-per-kilogram estimates for launches to LEO. Others require a simple calculation: dividing the total cost of a dedicated launch by the vehicle’s payload capacity to LEO. In those cases, the reported cost-per-kilogram figure is calculated by the *median* total launch cost and the *maximum* payload capacity.

In this data repository, the “number of successful orbital launches” includes all launches before December 31, 2019.

Definitions

According to the RAND Corporation, the **unit flyaway cost** “includes all direct and indirect manufacturing costs and their associated overhead plus recurring engineering, sustaining tooling, and quality control.”³ Unit flyaway cost often includes “[a]llowances or allocations to cover system and program management, software and other engineering changes and their associated test, and nonrecurring tooling, manufacturing, and engineering.”

A **dedicated launch**, also known as a “single-manifest” launch, is a launch in which the vehicle’s payload capacity is dedicated to one particular customer, as opposed to several customers sharing the available payload mass.⁴ Two or more customers sharing a launch is known as “ride-sharing.”

The **maximum payload capacity** to LEO for a space launch vehicle is simply the highest mass capacity reported by a launch provider. Often, the maximum payload capacity is calculated by assuming a relatively low-altitude circular orbit, such 185 km, and an inclination that corresponds to the latitude of one of the vehicle’s preferred spaceports. If the same space launch vehicle were to support a different mission to LEO, such as one that requires a higher altitude or inclination, the payload capacity would be reduced.

Although space launch vehicles are often described by their **payload mass class**—most often “Small,” “Medium,” and “Heavy”—there is no universally accepted definition for the boundaries between these classes.⁵ In this data repository, small-lift vehicles carry up to 2,000 kg to LEO, medium-lift vehicles carry between 2,000 and 20,000 kg to LEO, and heavy-lift vehicles carry more than 20,000 kg to LEO.

This interactive data repository is a product of the Andreas C. Dracopoulos iDeas Lab, the in-house digital, multimedia, and design agency at the Center for Strategic and International Studies.

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